

UNIVERSITY OF SOUTHERN CALIFORNIA

# EE 483: Digital Signal Processing

## Lecture 3: Properties of Discrete-Time Systems

*Lecturer*      Prof. Justin Haldar

*Collected by*      Gewenxin Yu

*Date*      September 1, 2026

---

1

## Last Time

Systems building blocks.

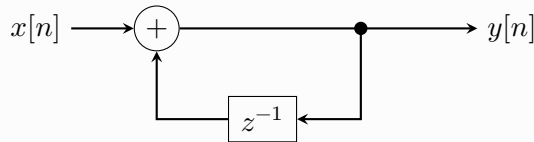
*Example (FIR Filter).*

A finite impulse response system, as introduced last lecture.

**Example (Accumulator System).**

$$y[n] = \sum_{\ell=-\infty}^n x[\ell], \quad y[n-1] = \sum_{\ell=-\infty}^{n-1} x[\ell] \implies y[n] - y[n-1] = x[n].$$

This can be calculated practically as  $y[n] = y[n-1] + x[n]$  (given some initial condition):



The impulse response is  $h[n] = H\{\delta[n]\} = u[n]$ . This system is **not FIR**; it is **IIR** (Infinite Impulse Response).

**Properties we will study:** [Linearity](#), [Time Invariance](#), [Causality](#), [Impulse Response](#), [Convolution](#), [Stability](#).

## 2 1 Linear Discrete-Time Systems

**Definition (Linearity).**

For arbitrary  $\alpha, \beta \in \mathbb{C}$  and arbitrary inputs  $x_1[n]$  and  $x_2[n]$ : if

$$y_1[n] = H\{x_1[n]\} \quad \left( x_1[n] \xrightarrow{H} y_1[n] \right)$$

and if

$$x_2[n] \xrightarrow{H} y_2[n],$$

then if

$$x[n] = \alpha x_1[n] + \beta x_2[n],$$

we must have

$$x[n] \xrightarrow{H} y[n] = \alpha y_1[n] + \beta y_2[n].$$

**Example (Accumulator is Linear).**

$$y[n] = \sum_{\ell=-\infty}^n x[\ell] \quad (\text{Accumulator})$$

$$x_1[n] \xrightarrow{H} y_1[n] = \sum_{\ell=-\infty}^n x_1[\ell]$$

$$x_2[n] \xrightarrow{H} y_2[n] = \sum_{\ell=-\infty}^n x_2[\ell]$$

If

$$x[n] = \alpha x_1[n] + \beta x_2[n],$$

then

$$\begin{aligned} x[n] \xrightarrow{H} y[n] &= \sum_{\ell=-\infty}^n (\alpha x_1[\ell] + \beta x_2[\ell]) \\ &= \alpha \sum_{\ell=-\infty}^n x_1[\ell] + \beta \sum_{\ell=-\infty}^n x_2[\ell] \\ &= \alpha y_1[n] + \beta y_2[n]. \end{aligned}$$

Therefore, the accumulator is linear.

## 3 2 Homogeneity for the Accumulator, via the Difference-Equation Form

**Example.**

$$y[n] = y[n - 1] + x[n] \quad (\text{accumulator})$$

Assume

$$x[n] \xrightarrow{H} y[n]$$

Let

$$x_1[n] = \alpha x[n] \xrightarrow{H} y_1[n]$$

Then

$$y_1[n] = y_1[n - 1] + \alpha x[n], \quad \alpha y[n] = \alpha y[n - 1] + \alpha x[n].$$

Subtracting,

$$\begin{aligned} y_1[n] - \alpha y[n] &= y_1[n - 1] - \alpha y[n - 1] \\ \Rightarrow g[n] &= g[n - 1], \quad g[n] \triangleq y_1[n] - \alpha y[n]. \end{aligned}$$

**Remark.**

$g[n]$  should be zero for all  $n$  for homogeneity to hold.

We assume **zero initial conditions (ICs)** (a restrictive definition; we will use something more general for LTI systems):

$$\begin{aligned} \exists M \in \mathbb{Z} \text{ s.t. } y[n] &= 0, \quad x[n] = 0, \quad y_1[n] = 0 \quad \forall n \leq M \\ \Rightarrow g[n] &= 0 \quad \forall n \leq M \\ \Rightarrow g[n] &= 0 \quad \forall n. \end{aligned}$$

So  $y_1[n] = \alpha y[n]$  for all  $n$ : **Homogeneity** ✓.

**Remark.**

Proving linearity is frequently harder than disproving linearity.

*Example (Median Filter: Homogeneous but not Additive).*

$$y[n] = \text{median}\{x[n-1], x[n], x[n+1]\}.$$

- Homogeneity? ✓
- Additivity? ✗

For single impulses  $x_1[n]$  and  $x_2[n]$  at different locations, the median filter outputs  $y_1[n]$  and  $y_2[n]$  that are each zero (or small); but for  $x_1[n] + x_2[n]$  (two nonzero samples), the median output  $y[n] \neq y_1[n] + y_2[n]$ . Hence **additivity fails**.

## 5 3 Time-Invariant Discrete-Time Systems

**Definition (Time-Invariance).**

For arbitrary  $x[n] \xrightarrow{H} y[n]$ : if  $x_1[n] = x[n - n_0]$  for arbitrary  $n_0 \in \mathbb{Z}$ , then  $x_1[n] \xrightarrow{H} y_1[n] = y[n - n_0]$ .

*Example (FIR Filter is Time-Invariant).*

$$y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] \quad (\text{FIR filter}).$$

Let  $x_1[n] = x[n - n_0]$ . Then

$$y_1[n] = \sum_{k=0}^{M-1} \alpha_k x_1[n-k] = \sum_{k=0}^{M-1} \alpha_k x[(n-k)-n_0] \stackrel{?}{=} y[n-n_0] \implies \text{time invariant!}$$

**Example (A System that is Not Time-Invariant).**

$$y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] + x[0].$$

For  $x[n] = \delta[n]$ :

$$y[n] = \begin{cases} \alpha_n + 1, & 0 \leq n \leq M-1 \\ 0 + 1, & \text{else} \end{cases}.$$

For  $x_1[n] = \delta[n-1]$ :

$$y_1[n] = \begin{cases} \alpha_{n-1} + 0, & 1 \leq n \leq M \\ 0 + 0, & \text{else} \end{cases}.$$

Since  $y_1[n] \neq y[n-1]$  (the added constant  $x[0]$  does not shift), **time invariant?**  
**Nope.**

The extra term always reads the input at index 0, not at a lag relative to  $n$ . Shifting the input by  $n_0$  therefore does not translate that contribution through time; the output is not a shifted copy of the original response.

## 4 Linear Time-Invariant (LTI) Systems

- Linear
- Time-invariant
- Easy to analyze, characterize, implement, and design.

### 4.1 Important Subclass: Linear Constant-Coefficient Difference Equations (LDEs)

An important subclass, with zero ICs:

$$\sum_{k=0}^{N-1} d_k y[n-k] = \sum_{q=0}^{M-1} p_q x[n-q].$$

Solving for  $y[n]$ :

$$y[n] = \sum_{k=1}^{N-1} \tilde{d}_k y[n-k] + \sum_{q=0}^{M-1} \tilde{p}_q x[n-q],$$

$$\left( \tilde{d}_k = -\frac{d_k}{d_0}, \quad \tilde{p}_q = \frac{p_q}{d_0}; \text{ assuming } d_0 \neq 0 \right).$$

7

*Examples: Accumulator (FIR Filter).*

$$y[n] = y[n-1] - x[n-1] + \rho(y[n-1] - x[n-1]),$$

where  $y[n]$  is the loan balance,  $y[n-1]$  the previous balance,  $x[n-1]$  the payment, and  $\rho$  the interest rate. Similar in spirit to the continuous-time ODE form

$$y(t) = \sum_{n=0}^N \alpha_n \frac{d^n x(t)}{dt^n} + \sum_{m=1}^M \beta_m \frac{d^m y(t)}{dt^m}.$$

## 5 Causal Systems

**Definition (Causality).**

The  $m$ th output sample  $y[m]$  depends only on input samples  $x[n]$  with  $n \leq m$ . This must be true for all possible inputs and all  $m$ .

*Examples.*

- $y[n] = \text{median}\{x[n-1], x[n], x[n+1]\}$ : **not causal**.
- $y[n] = y[n-1] + x[n]$ : causal, with an IC.
- $y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k]$ : **causal**.
- $y[n] = \sum_{k=0}^{M-1} \alpha_k x[n-k] + x[0]$ : **not causal**.

8

**Property (Causal Systems and Matching Inputs).**

For a causal system, if  $x_1[n] \xrightarrow{H} y_1[n]$  and  $x_2[n] \xrightarrow{H} y_2[n]$ , and if  $x_1[n] = x_2[n]$  for all  $n < N$ , and the ICs are the same, then  $y_1[n] = y_2[n]$  for all  $n < N$ .

## 6 Stability

**Definition (BIBO Stability).**

A system is **bounded-input bounded-output (BIBO) stable** if every bounded input yields a bounded output, i.e., if  $|x[n]| \leq B_x < \infty$  for all  $n$ , then  $|y[n]| \leq B_y < \infty$  for all  $n$ .

9

**Example (Moving Average is BIBO Stable).**

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k].$$

Assume

$$|x[n]| \leq B_x < \infty \quad \text{for all } n.$$

Then

$$\begin{aligned} |y[n]| &= \left| \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \right| \\ &\leq \frac{1}{M} \sum_{k=0}^{M-1} |x[n-k]| \\ &\leq \frac{1}{M} \sum_{k=0}^{M-1} B_x = B_x \implies \text{BIBO stable!} \end{aligned}$$

**Example (Accumulator is Not BIBO Stable).**

$y[n] = y[n-1] + x[n]$ , with IC  $y[n] = x[n] = 0$  for  $n < 0$ . Let  $x[n] = u[n]$  (bounded). Then  $y[n] = (n+1)u[n]$ , which is **not bounded**.

## 7 Impulse Response

**Definition (Impulse Response).**

The impulse response  $h[n]$  is the output of a discrete-time system to the input  $\delta[n]$ .

**Example (Accumulator).**

$$y[n] = \sum_{k=-\infty}^n x[k] \implies h[n] = \sum_{k=-\infty}^n \delta[k] = u[n].$$

10

**Fact.**

LTI systems with zero ICs are **completely characterized** by their impulse responses.

**Remark.**

A more general definition of zero ICs for LTI systems: if  $x[n] = 0$  for all  $n$ , then  $y[n] = H\{x[n]\} = 0$  for all  $n$ : “zero-input”  $\Rightarrow$  “zero-output.” (Non-zero ICs  $\Rightarrow$  not LTI.)

*Proof of fact:* next time.

## Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.5–4.7
- Monson H. Hayes, *Schaum’s Outline of Digital Signal Processing*, 2nd ed., Ch. 1.5, 2.4